

HandsMen Threads Salesforce Project Documentation

Prepared for Project Review

Salesforce Platform Developer Project

Data Model Automation Security Apex Testing

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1 Project Overview

HandsMen Threads is a Salesforce solution for a fashion business that manages customers, products, orders, inventory, and marketing campaigns. The project brings operational data into one platform and applies automation to improve customer communication, stock control, loyalty management, and order processing.

The implementation combines custom objects, custom fields, lookup and master detail relationships, formula fields, validation rules, profiles, roles, permission sets, email templates, record triggered flows, scheduled automation, Apex triggers, Batch Apex, and Scheduled Apex.

Main Business Outcomes

- Send an automatic confirmation message when an order is confirmed.
- Update customer loyalty status from purchase history.
- Notify the inventory team when stock falls below the defined threshold.
- Calculate order totals and deduct stock when confirmed orders are processed.
- Process inventory updates in bulk through scheduled Apex.

Technology Used

Area	Salesforce Feature
Data	Custom objects, custom fields, formula fields, relationships
Automation	Record triggered flows and scheduled flow
Development	Apex classes, triggers, handler pattern, Batch Apex, Scheduled Apex
Security	Profiles, roles, users, permission sets, object access
Communication	Classic email templates and email alerts

2 Objectives and Implementation Scope

2.1 Project Objectives

- Create a structured Salesforce data model for customers, products, orders, inventory, and campaigns.
- Maintain accurate customer, order, and stock information through UI level validation.
- Automate order confirmation, low stock alerts, and loyalty status updates.
- Apply role based access so teams can work only with relevant business records.
- Use Apex for order validation, total calculation, stock deduction, and bulk processing.

2.2 Current Implementation Scope

Area	Included Implementation	Review Evidence
Data model	Five custom objects with business fields and relationships	Object Manager and Schema Builder
Data quality	Formula fields and validation rules	Field and validation rule screens
Security	Profile, role hierarchy, users, permission set	Setup security screens
Automation	Order, stock, and loyalty flows	Flow Builder canvases
Development	Handler, triggers, batch, and scheduler	Apex class and trigger screens
Testing	Sample records, error messages, updates, and emails	Lightning records and test results

3 Requirement Analysis and Data Model

The data model separates master data from transaction data. Customer and product records store reusable business information. Orders capture transactions, inventory records track stock by product, and marketing campaigns connect promotional activity with customers.

Custom Object	Key Fields	Business Purpose
HandsMen Customer	Name, First Name, Last Name, Email, Phone, Loyalty Status, Total Purchases	Stores customer identity, contact, and loyalty details.
HandsMen Product	Name, SKU, Price, Stock Quantity	Stores the product catalogue and pricing.
HandsMen Order	Order Number, Status, Quantity, Total Amount, Customer, Product	Captures customer purchases and order status.
Inventory	Inventory Number, Warehouse, Stock Quantity, Product	Tracks stock levels for each product.
Marketing Campaign	Campaign Number, Start Date, End Date, Customer	Stores promotional campaigns and customer association.

SETUP > OBJECT MANAGER HandsMen Customer				
Details	Fields & Relationships 11 Items, Sorted by Field Label			
Fields & Relationships	Q, Quick Find New Deleted Fields Field Dependencies Set History Tracking			
Page Layouts	FIELD LABEL	FIELD NAME	DATA TYPE	CONTROLLING FIELD INDEXED
Lightning Record Pages	Created By	CreatedById	Lookup(User)	
Buttons, Links, and Actions	Email	Email__c	Email	
Compact Layouts	FirstName	FirstName__c	Text(20)	
Field Sets	FullName	FullName__c	Formula (Text)	
Object Limits	HandsMen Customer Name	Name	Text(80)	✓
Record Types	Last Modified By	LastModifiedById	Lookup(User)	
Related Lookup Filters	LastName	LastName__c	Text(20)	
Search Layouts	Loyalty Status	Loyalty_Status__c	Picklist	
List View Button Layout	Owner	OwnerId	Lookup(User,Group)	✓
Restriction Rules	Phone	Phone__c	Phone	
Scoping Rules	Total Purchases	Total_Purchases__c	Number(18, 0)	
Object Access				
Field Access				

Figure 1 HandsMen Customer object fields including email phone loyalty status and total purchases

Setup Home Object Manager Search Setup				
SETUP > OBJECT MANAGER HandsMen Product				
Details Fields & Relationships Page Layouts Lightning Record Pages Buttons, Links, and Actions Compact Layouts Field Sets Object Limits Record Types Related Lookup Filters Search Layouts List View Button Layout Restriction Rules Scooina Rules	Fields & Relationships 8 Items, Sorted by Field Label Quick Find New Deleted Fields Field Dependencies Set History Tracking			
	FIELD LABEL	FIELD NAME	DATA TYPE	CONTROLLING FIELD INDEXED
	Created By	CreatedById	Lookup(User)	
	HandsMen Product Name	Name	Text(80)	✓
	Last Modified By	LastModifiedById	Lookup(User)	
	Order	Order__c	Lookup(HandsMen Order)	✓
	Owner	OwnerId	Lookup(User,Group)	✓
	Price	Price__c	Currency(18, 0)	
	SKU	SKU__c	Text(60)	
	Stock Quantity	Stock_Quantity__c	Number(18, 0)	

Figure 2 HandsMen Product object fields including SKU price and stock quantity

SETUP > OBJECT MANAGER HandsMen Order				
Details Fields & Relationships Page Layouts Lightning Record Pages Buttons, Links, and Actions Compact Layouts Field Sets Object Limits Record Types Related Lookup Filters Search Layouts List View Button Layout Restriction Rules Scoping Rules Object Access Field Access	Fields & Relationships 10 Items, Sorted by Field Label Quick Find New Deleted Fields Field Dependencies Set History Tracking			
	FIELD LABEL	FIELD NAME	DATA TYPE	CONTROLLING FIELD INDEXED
	Created By	CreatedById	Lookup(User)	
	Customer	Customer__c	Lookup(HandsMen Customer)	✓
	Customer email	Customer_email__c	Email	
	HandsMen OrderNumber	Name	Auto Number	✓
	Last Modified By	LastModifiedById	Lookup(User)	
	Owner	OwnerId	Lookup(User,Group)	✓
	Product	Product__c	Lookup(HandsMen Product)	✓
	Quantity	Quantity__c	Number(18, 0)	
	Status	Status__c	Picklist	
	Total Amount	Total_Amount__c	Number(18, 0)	

Figure 3 HandsMen Order object fields and relationships including status quantity total amount customer and product

SETUP > OBJECT MANAGER				
Inventory				
Details	Fields & Relationships			
Fields & Relationships	7 Items, Sorted by Field Label			
Page Layouts	Q, Quick Find New Deleted Fields Field Dependencies Set History Tracking			
Lightning Record Pages	FIELD LABEL	FIELD NAME	DATA TYPE	CONTROLLING FIELD INDEXED
Buttons, Links, and Actions	Created By	CreatedById	Lookup(User)	
Compact Layouts	Inventory Number	Name	Auto Number	✓
Field Sets	Last Modified By	LastModifiedById	Lookup(User)	
Object Limits	Product	Product__c	Master-Detail(HandsMen Product)	✓
Record Types	Stock Quantity	Stock_Quantity__c	Number(18, 0)	
Related Lookup Filters	Stock Status	Stock_Status__c	Formula (Text)	
Search Layouts	Warehouse	Warehouse__c	Text(60)	
List View Button Layout				
Restriction Rules				
Scoping Rules				

Figure 4 Inventory object fields and relationship including warehouse stock quantity and related product

SETUP > OBJECT MANAGER				
Marketing Campaign				
Details	Fields & Relationships			
Fields & Relationships	7 Items, Sorted by Field Label			
Page Layouts	Q, Quick Find New Deleted Fields Field Dependencies Set History Tracking			
Lightning Record Pages	FIELD LABEL	FIELD NAME	DATA TYPE	CONTROLLING FIELD INDEXED
Buttons, Links, and Actions	Created By	CreatedById	Lookup(User)	
Compact Layouts	End Date	End_Date__c	Date	
Field Sets	HandsMen Customer	HandsMen_Customer__c	Lookup(HandsMen Customer)	✓
Object Limits	Last Modified By	LastModifiedById	Lookup(User)	
Record Types	Marketing Campaign Number	Name	Auto Number	✓
Related Lookup Filters	Owner	OwnerId	Lookup(User,Group)	✓
Search Layouts	Start Date	Start_Date__c	Date	
List View Button Layout				
Restriction Rules				
Scoping Rules				

Figure 6 Marketing Campaign object fields and relationship including campaign dates and related customer

3.1 Relationships and Important Fields

Child Object	Relationship	Related Object	Purpose
Marketing Campaign	Lookup	HandsMen Customer	Associates a campaign with a customer.
HandsMen Product	Lookup	HandsMen Order	Connects product information to an order as documented.
HandsMen Order	Lookup	HandsMen Customer	Links each order to the purchasing customer.
Inventory	Master Detail	HandsMen Product	Makes inventory dependent on its product.

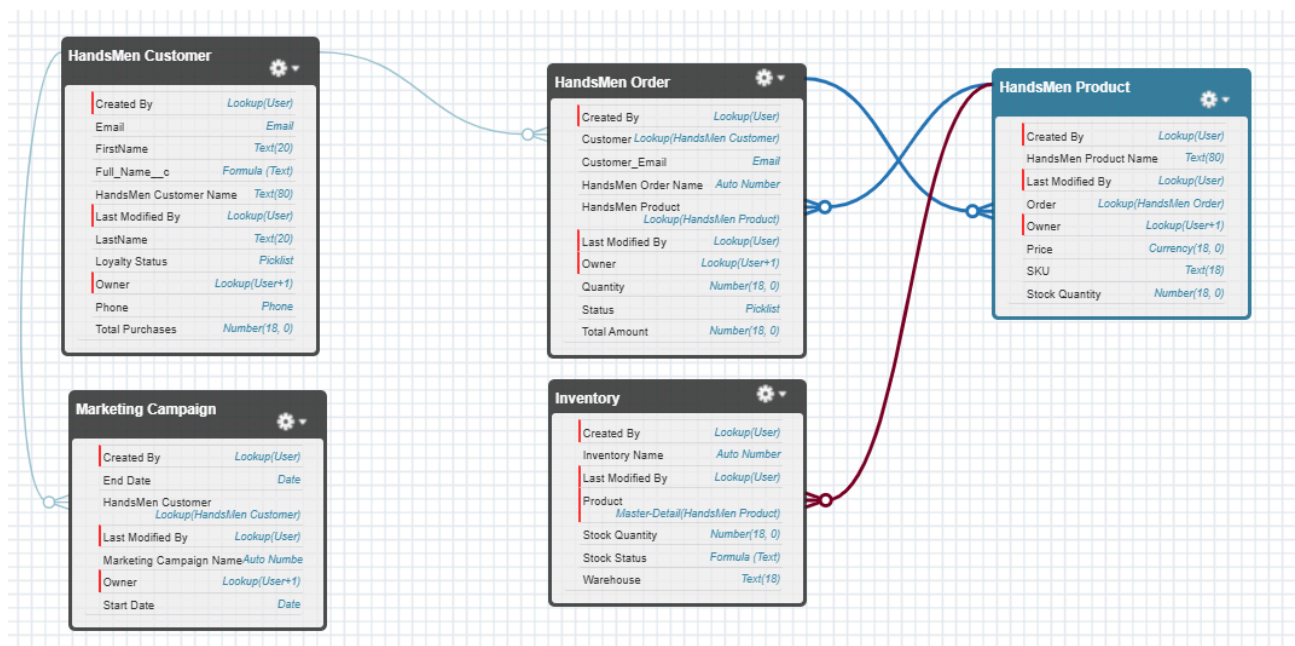


Figure 2 Schema Builder Data Model

4 Salesforce Configuration and Data Security

The project uses custom tabs and a dedicated navigation workspace so users can open customer, product, order, inventory, and campaign records from Lightning Experience. Security is controlled through profiles, roles, and a permission set.

Role	Intended Access
Sales Manager	Full access to customer and order records.
Inventory Manager	Read and edit access to inventory and product records.
Marketing Manager or Team	Read customer records and manage marketing campaign records.

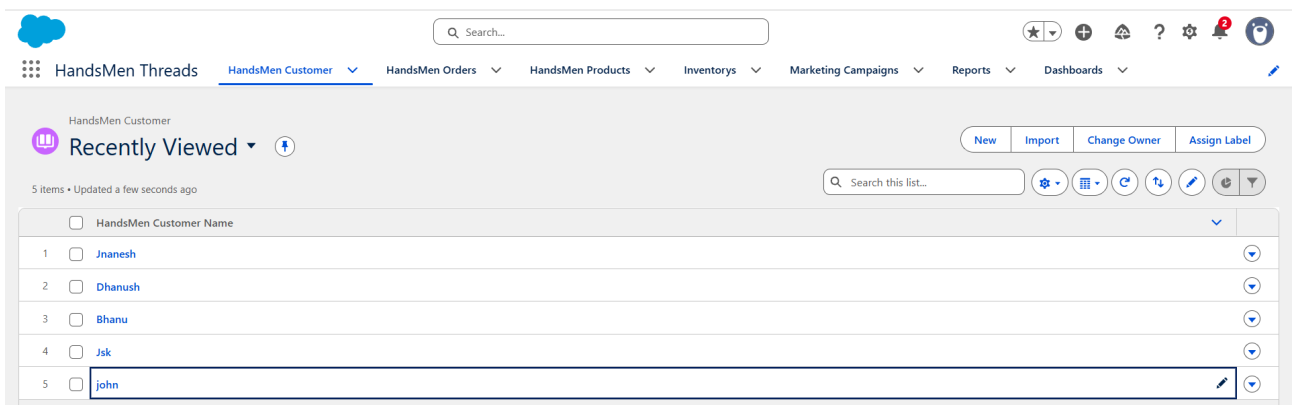


Figure 4 HandsMen Lightning App and Navigation



Figure 5 Role Hierarchy

4.1 Profiles Users and Permission Sets

Security Item	Configuration	Purpose
Profile	Platform 1 cloned from Standard User	Provides the base Salesforce Platform access.
User	Sales user assigned to Sales role	Supports customer and order activities.
User	Inventory user assigned to Inventory role	Supports product and inventory activities.
Permission Set	Permission Platform 1	Adds create, read, edit, and delete access for selected project objects.


SETUP
Permission Sets

Permission Set
sales permission set

Clone
Edit Properties
Manage Assignments
View Summary

[Permission Set Overview](#) > [Object Settings](#) ▼ **HandsMen Customer** ▼

HandsMen Customer Edit

Tab Settings

Available	Visible
☐	☐ i

Object Permissions

Permission Name	Enabled
Read	☒
Create	☒
Edit	☒
Delete	☒
View All Records	☐
Modify All Records	☐
View All Fields	☐

Figure 1 HandsMen Customer permissions configured in Permission Platform 1


SETUP

Permission Sets

Permission Set

sales permission set

✕
i
Clone
Edit Properties
Manage Assignments
View Summary

[Permission Set Overview](#) >
[Object Settings](#)
▼
HandsMen Orders
▼

HandsMen Orders
Edit

Tab Settings

Available	Visible
☐	☐ i

Object Permissions

Permission Name	Enabled
Read	☒
Create	☒
Edit	☒
Delete	☒
View All Records	☐
Modify All Records	☐
View All Fields	☐

Figure 2 HandsMen Order permissions configured in Permission Platform 1


SETUP

Permission Sets

Permission Set

sales permission set

✕
i
Clone
Edit Properties
Manage Assignments
View Summary

[Permission Set Overview](#) >
[Object Settings](#)
▼
HandsMen Products
▼

HandsMen Products
Edit

Tab Settings

Available	Visible
☐	☐ i

Object Permissions

Permission Name	Enabled
Read	☒
Create	☒
Edit	☒
Delete	☒
View All Records	☐
Modify All Records	☐
View All Fields	☐

Figure 3 HandsMen Products permissions configured in Permission Platform 1

SETUP

Users

All Users

On this page you can create, view, and manage users.

To get more licenses, use the Your Account app. [Let's Go](#)

View: All Users [Edit](#) [Create New View](#)

A B C D E F G H I J K L M N O P Q R S T U V W X Y Z Other **All**

New User

Reset Password(s)

Add Multiple Users

☐ Action	Full Name +	Alias	Username	Role	Active	Profile
☐ Edit	Chatter Expert	Chatter	chatty.00dg80000h05kteaz.n6dacketlllll@chatter.salesforce.com		☒	Chatter Free User
☐ Edit	EPIC_OrgFarm	OEPIC	epic.f4bb6d103251@orgfarm.salesforce.com		☒	System Administrator
☐ Edit	Mikaelson Danny	dmika	jnanesh2.sunkara@gmail.com	Marketing	☒	Platform 1
☐ Edit	Mikaelson Kol	kmika	jnanesh.sunkara1@gmail.com	Inventory	☐	Platform 1
☐ Edit	Mikaelson Niklaus	nmika	jnanesh.sunkara@gmail.com	Sales	☒	Read Only
☐ Edit	Sunkara Jnanesh	jna	jnaneshsunkara.ai.a8e9fc39dd9@agentforce.com		☒	System Administrator
☐ Edit	User Integration	integ	integration@00dg80000h05kteaz.com		☒	Analytics Cloud Integration User
☐ Edit	User Security	sec	insightssecurity@00dg80000h05kteaz.com		☒	Analytics Cloud Security User

New User

Reset Password(s)

Add Multiple Users

Figure 4 Project Users and Assigned Roles

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5 Data Quality and Business Rules

5.1 Formula Fields

Object	Formula Field	Return Type	Formula
HandsMen Customer	Full Name	Text	FirstName__c + " " + LastName__c
Inventory	Stock Status	Text	IF(Stock_Quantity__c > 10, "Available", "Low Stock")

5.2 Validation Rules

Object	Rule	Condition	Expected Message
HandsMen Order	Total Amount	Total_Amount__c <= 0	Please enter a correct amount.
Inventory	Stock Quantity	Stock_Quantity__c <= 0	Inventory count cannot be less than or equal to zero.
HandsMen Customer	Email	NOT(CONTAINS(Email__c, "@gmail.com"))	Please enter a correct Gmail address.

The screenshot shows the Salesforce Setup interface for configuring a validation rule on the 'HandsMen Order' object. The 'Error Condition Formula' is set to 'Total_Amount__c <= 0'. The 'Check Syntax' button shows 'No errors found'.

Setup > OBJECT MANAGER
HandsMen Order

Details
Fields & Relationships
Page Layouts
Lightning Record Pages
Buttons, Links, and Actions
Compact Layouts
Field Sets
Object Limits
Record Types
Related Lookup Filters
Search Layouts

ACTIVE ☒
Description

Error Condition Formula

Example: `Discount_Percent__c > 0.30` [More Examples...](#)
Display an error if Discount is more than 30%
If this formula expression is **true**, display the text defined in the Error Message area

`Total_Amount__c <= 0`

No errors found

Functions
-- All Function Categories --
ABS
ACOS
ADDMONTHS
AND
ASCII
ASIN

ABS(number)
Returns the absolute value of a number, a number without its sign
[Help on this function](#)

Figure 1 Total Amount field configuration and successful syntax validation

Setup Home Object Manager

SETUP > OBJECT MANAGER

HandsMen Customer

Details
Fields & Relationships
Page Layouts
Lightning Record Pages
Buttons, Links, and Actions
Compact Layouts
Field Sets
Object Limits
Record Types
Related Lookup Filters
Search Layouts

Rule Name:

Active: ☒

Description:

Error Condition Formula

Example: `Discount_Percent__c > 0.30` [More Examples...](#)
Display an error if Discount is more than 30%
If this formula expression is **true**, display the text defined in the Error Message area

`NOT CONTAINS ( Email__c , "@gmail.com")`

No errors found

Functions

-- All Function Categories --

ABS

ACOS

ADDMONTHS

AND

ASCII

ASIN

ABS(number)

Returns the absolute value of a number, a number without its sign

[Help on this function](#)

Figure 2 Email formula field configuration and successful syntax validation

Setup Home Object Manager

SETUP > OBJECT MANAGER

Inventory

Details
Fields & Relationships
Page Layouts
Lightning Record Pages
Buttons, Links, and Actions
Compact Layouts
Field Sets
Object Limits
Record Types
Related Lookup Filters
Search Layouts

Rule Name:

Active: ☒

Description:

Error Condition Formula

Example: `Discount_Percent__c > 0.30` [More Examples...](#)
Display an error if Discount is more than 30%
If this formula expression is **true**, display the text defined in the Error Message area

`Stock_Quantity__c <= 0`

No errors found

Functions

-- All Function Categories --

ABS

ACOS

ADDMONTHS

AND

ASCII

ASIN

ABS(number)

Returns the absolute value of a number, a number without its sign

[Help on this function](#)

Figure 3 Stock Quantity configuration and successful syntax validation

New HandsMen Customer

* = Required Information

Information

* HandsMen Customer Name Michael

Email Miky23
Enter a valid email address, such as name@email.com.

Phone

Loyalty Status --None--

FirstName

LastName

Total Purchases

Owner Jnanesh Sunkara

⚠ We hit a snag.
Review the following fields
• Email

⚠ Cancel Save & New Save

Figure 3 Customer email validation rule preventing an invalid email from being saved

6 Email Templates and Process Automation

6.1 Email Templates and Alerts

Template	Type	Used For
Order Confirmation	HTML	Sent when an order changes to Confirmed.
Low Stock Alert	Text	Sent to the inventory team when stock is below five.
Loyalty Program Email	HTML	Supports customer loyalty communication.

SETUP Classic Email Templates						
Edit Del	Appointment for Unauthenticated User using Appointment Types - For Amazon Chime	Custom	✓	Email template for confirmation of an appointment when appointments are scheduled using appointment types with Amazon Chime.	sfdcadmin	9/2/2026
Edit Del	Appointment for Unauthenticated User using Appointment Types - For third party	Custom	✓	Email template for confirmation of an appointment when appointments are scheduled using appointment types with third party video applications.	sfdcadmin	9/2/2026
Edit Del	Appointment for Unauthenticated User using Engagement Channels For Amazon Chime	Custom	✓	Email template for confirmation of an appointment when appointments are scheduled using engagement channels with Amazon Chime.	sfdcadmin	9/2/2026
Edit Del	Appointment for Unauthenticated User using Engagement Channels For third party	Custom	✓	Email template for confirmation of an appointment when appointments are scheduled using engagement channels with third party video applications.	sfdcadmin	9/2/2026
Edit Del	Canceled Service Appointment Confirmation Email	Custom	✓	Email Template to confirm canceling of a service appointment.	sfdcadmin	9/2/2026
Edit Del	Commerce Reorder Portal: Invitation	Custom	✓	Invite a contact to a Commerce Reorder Portal.	auto@oos	9/2/2026
Edit Del	Group Service Appointments Enrollment Confirmation Email	Custom	✓	Email Template to confirm enrollment of an attendee to a Group service appointment.	sfdcadmin	9/2/2026
Edit Del	Low Stock Alert	Text	✓		ina	9/12/2026
Edit Del	Loyalty Program Email	HTML	✓		ina	9/15/2026
Edit Del	Marketing Product Inquiry Response	Text	✓	Standard email response to website product inquiries	OEPIC	9/2/2026
Edit Del	Order Confirmation Email	HTML	✓		ina	9/12/2026
Edit Del	Rescheduled Service Appointment Confirmation Email	Custom	✓	Email Template to confirm rescheduling of a service appointment.	sfdcadmin	9/2/2026
Edit Del	Sales New Customer Email	Text	✓	Email to new customers	OEPIC	9/2/2026
Edit Del	Scheduled Service Appointment Confirmation Email	Custom	✓	Email Template to confirm scheduling of a service appointment.	sfdcadmin	9/2/2026
Edit Del	Scheduler Payments Payment Reminder for Service Appointment Email	Custom	✓	Email Template to remind customers to pay for their service appointment.	sfdcadmin	9/2/2026
Edit Del	Scheduler Payments Service Appointment Cancellation Email	Custom	✓	Email Template to confirm the cancellation of a paid service appointment.	sfdcadmin	9/2/2026
Edit Del	Scheduler Payments Service Appointment Confirmation Email	Custom	☐	Email Template to confirm scheduling of a paid service appointment.	sfdcadmin	9/2/2026
Edit Del	Scheduler Payments Service Appointment Confirmation Email for Guest Users	Custom	☐	Email Template to confirm scheduling of a paid service appointment for guest users.	sfdcadmin	9/2/2026
Edit Del	Scheduler Payments Service Appointment Rescheduled Email - Authenticated Users	Custom	✓	Email Template to confirm the rescheduling of a paid service appointment for authenticated users.	sfdcadmin	9/2/2026
Edit Del	Scheduler Payments Service Appointment Rescheduled Email - Guest Users	Custom	✓	Email Template to confirm the rescheduling of a paid service appointment for guest users.	sfdcadmin	9/2/2026
Edit Del	SUPPORT Self-Service New Comment Notification (SAMPLE)	Text	✓	Sample email template that can be sent to your Self-Service customers to notify them a public comment has been added to their case.	OEPIC	9/2/2026
Edit Del	SUPPORT Self-Service New User Login Information (SAMPLE)	Text	✓	Notification of login and password to new Self-Service user	OEPIC	9/2/2026
Edit Del	SUPPORT Self-Service Reset Password (SAMPLE)	Text	✓	Notification of new password when Self-Service password is reset	OEPIC	9/2/2026
Edit Del	Support Case Assignment Notification	Text	✓	Notification to rep when case is auto-assigned	OEPIC	9/2/2026
Edit Del	Support Case Created (Phone Inquiries)	Text	✓	Notification to customer about case created through phone call	OEPIC	9/2/2026

Figure 10 Email Templates List

SETUP Email Alerts				
All Email Alerts				
Email alerts are used to send emails from a flow or other automation.				
View: All Email Alerts Create New View				
A B C D E F G H I J K L M N O P Q R S T U V W X Y Z Other				
New Email Alert				
Action	Description ↑	Email Template Name	Object	Last Modified Date
Edit Del	Low Stock Alert	Low Stock Alert	Inventory	9/11/2026
Edit Del	Loyalty program email	Loyalty Program Email	HandsMen Customer	9/11/2026
Edit Del	Order Confirmation Email Alert	Order Confirmation Email	HandsMen Order	9/11/2026

Figure 11 Order Confirmation Email Alert

6.2 Salesforce Flows

Flow	Type	Start Condition	Action
Order Confirmation Flow	Record Triggered	Order status changes to Confirmed	Sends the order confirmation email alert.
Stock Alert Flow	Record Triggered	Inventory stock quantity is below 5	Sends a low stock email alert.
Loyalty Status Update Flow	Schedule Triggered	Runs daily	Sets Gold, Silver, or Bronze from Total Purchases.

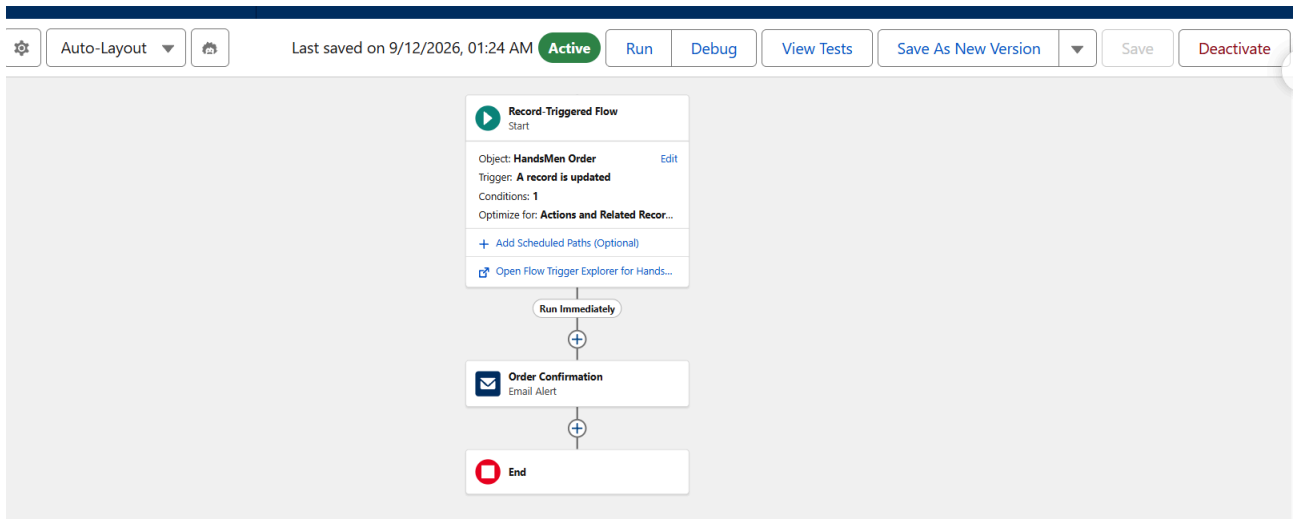


Figure 12 Order Confirmation Flow

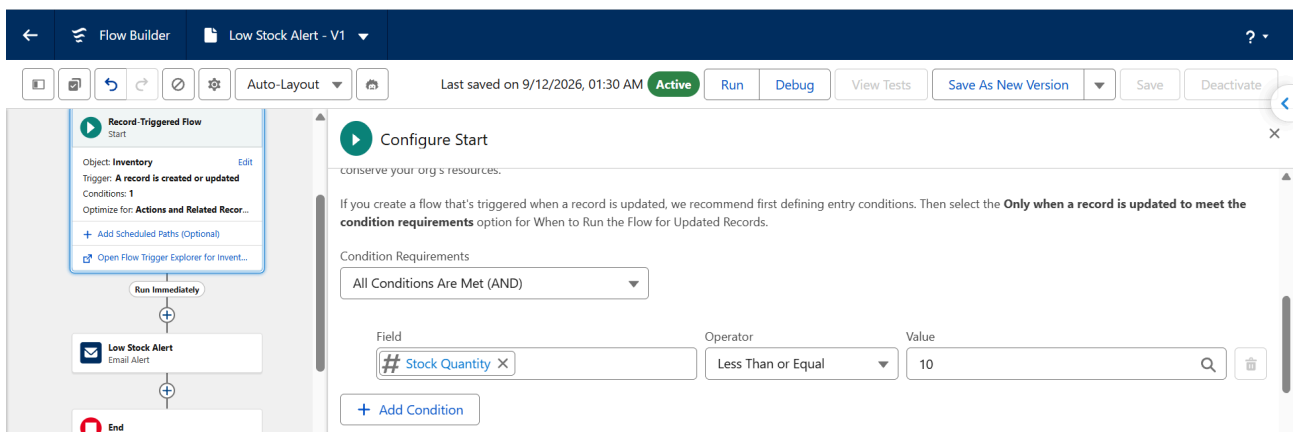


Figure 13 Stock Alert Flow

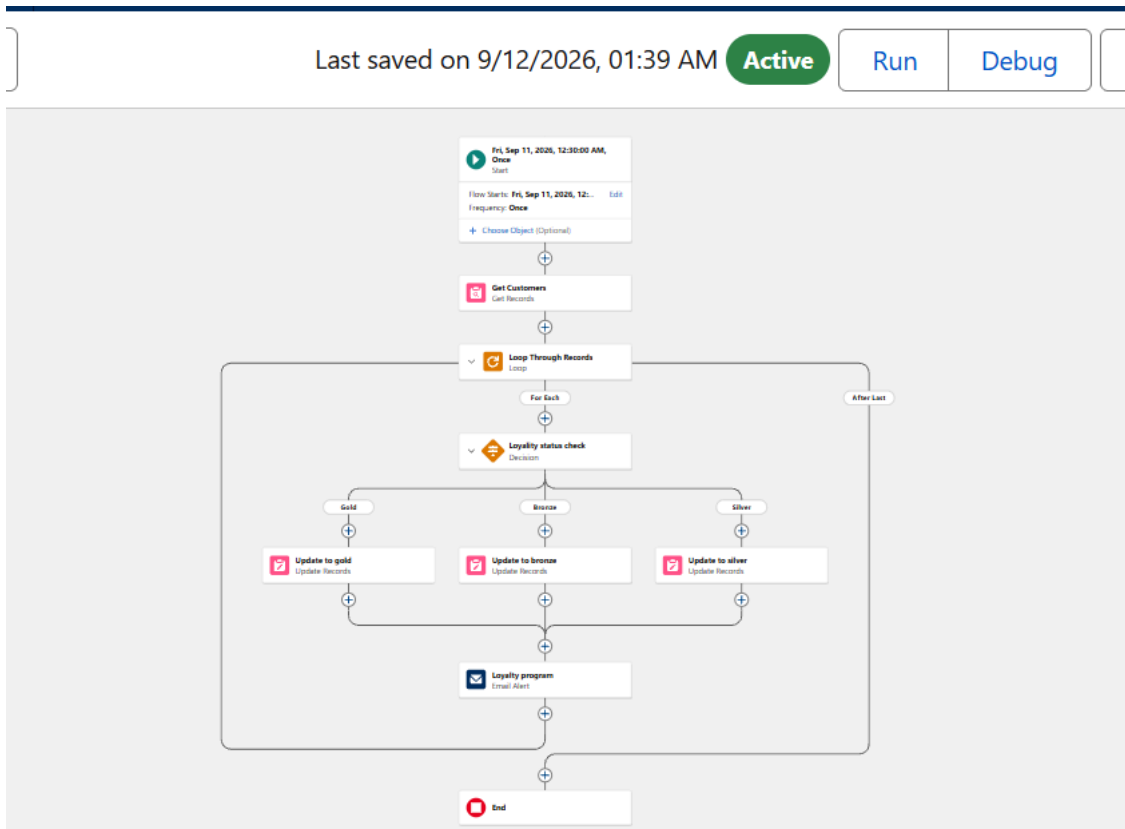


Figure 14 Loyalty Status Update Flow

7 Apex Triggers and Handler Logic

Apex is used where record processing requires reusable logic and transaction level enforcement. The order trigger delegates status based quantity validation to OrderTriggerHandler so the trigger remains small and the rule can be maintained in one class.

Status	Quantity Rule	Outcome
Confirmed	Quantity must be greater than 500	The record is blocked when the quantity is missing or too low.
Pending	Quantity must be greater than 200	The record is blocked when the quantity is missing or too low.
Rejection	Quantity must equal 0	Rejected orders cannot contain a positive quantity.

```

trigger OrderTrigger on HandsMen_Order__c (before insert, before update) {
    OrderTriggerHandler.validateOrderQuantity(Triiger.new);
}

```

```

public class OrderTriggerHandler {

    public static void validateOrderQuantity(List<HandsMen_Order__c> orderList) {

        for (HandsMen_Order__c order : orderList) {

            if (order.Status__c == 'Confirmed') {

                if (order.Quantity__c == null || order.Quantity__c <= 500) {

                    order.Quantity__c.addError('For Status "Confirmed", Quantity must be more than 500.');

```

Figure 15 OrderTriggerHandler Apex Class

```

trigger OrderTrigger on HandsMen_Order__c (before insert, before update) {

    if (Trigger.isBefore && (Trigger.isInsert || Trigger.isUpdate)) {

        OrderTriggerHandler.validateOrderQuantity(Trigger.new);

    }

}

```

Figure 16 OrderTrigger Apex Trigger

7.1 Order Total and Stock Deduction Triggers

Apex Artifact	Event	Responsibility
Order Total Trigger	Before insert and before update	Reads the related product price and sets Total Amount to Quantity multiplied by Price.
Stock Deduction Trigger	After insert and after update	When an order becomes Confirmed, totals ordered quantity by product and reduces Inventory Stock Quantity.

The stock deduction design collects product identifiers in a set, aggregates quantities in a map, queries matching inventory records once, and performs one bulk update. This avoids database operations inside the record loop.

```

trigger OrderTotalTrigger on HandsMen_Order__c (before insert, before update) {
    Set<Id> productIds = new Set<Id>();

    for (HandsMen_Order__c order : Trigger.new) {
        if (order.HandsMen_Product__c != null) {
            productIds.add(order.HandsMen_Product__c);
        }
    }

    Map<Id, HandsMen_Product__c> productMap = new Map<Id, HandsMen_Product__c>(
        [SELECT Id, Price__c FROM HandsMen_Product__c WHERE Id IN :productIds]
    );

    for (HandsMen_Order__c order : Trigger.new) {
        if (order.HandsMen_Product__c != null && productMap.containsKey(order.HandsMen_Product__c)) {
            HandsMen_Product__c product = productMap.get(order.HandsMen_Product__c);
            if (order.Quantity__c != null) {
                order.Total_Amount__c = order.Quantity__c * product.Price__c;
            }
        }
    }
}

```

Figure 17 Order Total Trigger

```

trigger StockDeductionTrigger on HandsMen_Order__c (after insert, after update) {
    Set<Id> productIds = new Set<Id>();
    for (HandsMen_Order__c order : Trigger.new) {
        Boolean shouldDeduct = false;

        if (order.Status__c == 'Confirmed') {
            if (Trigger.isInsert) {
                shouldDeduct = true;
            }
            else if (Trigger.isUpdate) {
                HandsMen_Order__c oldOrder = Trigger.oldMap.get(order.Id);

                if (oldOrder.Status__c != 'Confirmed') {
                    shouldDeduct = true;
                }
            }
        }

        if (shouldDeduct && order.HandsMen_Product__c != null) {
            productIds.add(order.HandsMen_Product__c);
        }
    }

    if (productIds.isEmpty()) {
        return;
    }

    // Deduct stock
    Map<Id, Inventory__c> inventoryMap = new Map<Id, Inventory__c>();
    for (Inventory__c inv : [
        SELECT Id, Stock_Quantity__c, Product__c
        FROM Inventory__c
        WHERE Product__c IN :productIds
    ]) {
        inventoryMap.put(inv.Product__c, inv);
    }

    List<Inventory__c> inventoriesToUpdate = new List<Inventory__c>();

    // Deduct stock
    for (HandsMen_Order__c order : Trigger.new) {
        Boolean shouldDeduct = false;

        if (order.Status__c == 'Confirmed') {
            if (Trigger.isInsert) {
                shouldDeduct = true;
            }
            else if (Trigger.isUpdate) {
                HandsMen_Order__c oldOrder = Trigger.oldMap.get(order.Id);

                if (oldOrder.Status__c != 'Confirmed') {
                    shouldDeduct = true;
                }
            }
        }

        if (shouldDeduct &&
            order.HandsMen_Product__c != null &&
            order.Quantity__c != null) {

            Inventory__c existingInventory =
                inventoryMap.get(order.HandsMen_Product__c);

            if (existingInventory != null) {

                Decimal currentStock =
                    existingInventory.Stock_Quantity__c == null
                    ? 0
                    : existingInventory.Stock_Quantity__c;

                Decimal newStock =
                    currentStock - order.Quantity__c;

                inventoriesToUpdate.add(
                    new Inventory__c(
                        Id = existingInventory.Id,
                        Stock_Quantity__c = newStock
                    )
                );
            }
        }
    }

    if (!inventoriesToUpdate.isEmpty()) {
        update inventoriesToUpdate;
    }
}

```

Figure 18 Stock Deduction Trigger

8 Batch Apex and Scheduled Processing

InventoryBatchJob implements Database.Batchable and Schedulable. It identifies low stock products, applies the restocking logic in manageable batches, and can be scheduled for automatic execution.

Method or Process	Responsibility	Expected Evidence
start	Queries products that meet the low stock condition.	SOQL query and selected fields.
execute	Adds the restock quantity and updates records in bulk.	Loop, update list, and DML statement.
finish	Records that inventory synchronization completed.	Completion debug statement.
Schedulable execute	Starts the batch with a scope of 200.	Database executeBatch call.
Scheduled job	Runs Daily Inventory Sync automatically.	Scheduled Jobs entry and next run time.

```
global class InventoryBatchJob implements Database.Batchable<SObject>, Schedulable {

    global Database.QueryLocator start(Database.BatchableContext BC) {

        return Database.getQueryLocator(

            'SELECT Id, Stock_Quantity__c FROM Product__c WHERE Stock_Quantity__c < 10'

        );

    }

    global void execute(Database.BatchableContext BC, List<SObject> records) {

        List<HandsMen_Product__c> productsToUpdate = new List<HandsMen_Product__c>();

        // Cast SObject list to Product__c list

        for (SObject record : records) {

            HandsMen_Product__c product = (HandsMen_Product__c) record;

            product.Stock_Quantity__c += 50; // Restock logic

            productsToUpdate.add(product);

        }

        28
        29 if (!productsToUpdate.isEmpty()) {
        30
        31 try {
        32
        33 update productsToUpdate;
        34
        35 } catch (DmlException e) {
        36
        37 System.debug('Error updating inventory: ' + e.getMessage());
        38
        39 }
        40
        41 }
        42
        43 }
        44
        45 global void finish(Database.BatchableContext BC) {
        46
        47 System.debug('Inventory Sync Completed');
        48
        49 }
        50
        51 // Scheduler Method
        52
        53 global void execute(SchedulableContext SC) {
        54
        55 InventoryBatchJob batchJob = new InventoryBatchJob();
        56
        57 Database.executeBatch(batchJob, 200);
        58
        59 }
        60
    }
```

Figure 19 InventoryBatchJob Apex Class



SETUP

Scheduled Jobs

All Scheduled Jobs

[Help for this Page](#)

The All Scheduled Jobs page lists all of the jobs scheduled by your users. Multiple job types may display on this page. You can delete scheduled jobs if you have the permission to do so.

**Percentage of Scheduled Jobs Used: 1%**

You have currently used 1 scheduled Apex jobs out of an allowed organization limit of 100 active or scheduled jobs. To learn about how this limit is calculated and what contributes to it see the [Lightning Platform Apex Limits](#) topic.

View: [All Scheduled Jobs](#) [Create New View](#)

[A](#) [B](#) [C](#) [D](#) [E](#) [F](#) [G](#) [H](#) [I](#) [J](#) [K](#) [L](#) [M](#) [N](#) [O](#) [P](#) [Q](#) [R](#) [S](#) [T](#) [U](#) [V](#) [W](#) [X](#) [Y](#) [Z](#) [Other](#) **All**

Schedule Apex							
Action	Job Name ↑	Submitted By	Submitted	Started	Next Scheduled Run	Type	Cron Trigger ID
Manage Del Pause Job	Daily Inventory Sync	Sunkara, Jnanesh	9/19/2026, 1:43 AM		9/20/2026, 12:00 AM	Scheduled Apex	08eg800000JXnlL

Figure 20 Scheduled Inventory Job

9 User Interface and Sample Records

Sample records should demonstrate that the configured fields, relationships, and automation are available to business users in Lightning Experience. Use realistic but non-sensitive data and keep the browser zoom between 80 and 100 percent so labels and values remain readable.

Record	Recommended Fields to Show
HandsMen Customer	Full name, email, phone, loyalty status, total purchases
HandsMen Product	Product name, SKU, price, stock quantity
HandsMen Order	Order number, customer, product, status, quantity, total amount
Inventory	Inventory number, product, warehouse, stock quantity, stock status
Marketing Campaign	Campaign number or name, dates, related customer

The screenshot displays the Salesforce Lightning Experience interface for 'HandsMen Threads'. At the top, there is a navigation bar with a search bar and tabs for 'HandsMen Customer', 'HandsMen Orders', 'HandsMen Products', 'Inventories', and 'Marketing Campaigns'. The 'HandsMen Customer' tab is active, showing a record for a customer named 'john'. The record details are displayed in a two-column layout. The left column contains fields for 'HandsMen Customer Name' (john), 'Email' (jnanesh.sunkara@gmail.com), 'Phone', 'Loyalty Status' (Bronze), 'FirstName', 'LastName', 'FullName', 'Total Purchases' (500), and 'Created By' (Jnanesh Sunkara, 9/12/2026, 4:42 AM). The right column contains fields for 'Owner' (Jnanesh Sunkara) and 'Last Modified By' (OrgFarm EPIC, 9/15/2026, 3:31 AM). Each field has an edit icon (pencil) next to it.

HandsMen Customer Name	john	Owner	Jnanesh Sunkara
Email	jnanesh.sunkara@gmail.com		
Phone			
Loyalty Status	Bronze		
FirstName			
LastName			
FullName			
Total Purchases	500		
Created By	Jnanesh Sunkara · 9/12/2026, 4:42 AM	Last Modified By	OrgFarm EPIC · 9/15/2026, 3:31 AM

Figure 21 Customer and Product Sample Records



O-0007

Details

O-0007



Confirmed

1,000

3,000

jnanesh.sunkara@gmail.com

T-shirt

 Jnanesh Sunkara, 9/19/2026, 1:38 AM

 Jnanesh Sunkara

 Jnanes Sunkara, 9/19/2026, 1:38 AM

Figure 22 Order and Inventory Sample Records

10 Functional Testing Evidence

Functional testing should show both configuration and business outcomes. Capture the action and its result, and use before and after screenshots where a value changes.

Test Case	Action	Expected Result	Evidence
Order quantity validation	Save a status and quantity combination that violates the handler rule.	Save is blocked with the Apex error.	Error on Order page
Order total	Create or edit an order with product and quantity.	Total Amount equals Quantity multiplied by Price.	Order values
Stock deduction	Change an order from Pending to Confirmed.	Inventory decreases once by the ordered quantity.	Before and after stock
Order confirmation	Confirm an order.	Customer receives the confirmation email.	Email inbox
Low stock alert	Set stock below 5.	Inventory manager receives an alert.	Email inbox
Loyalty update	Run or debug the scheduled flow with purchase values.	Customer loyalty changes to Gold, Silver, or Bronze.	Flow debug and record
Batch processing	Execute InventoryBatchJob.	Low stock records are processed and the job completes.	Apex Jobs and record

The screenshot displays the 'HandsMen Order' form in the 'HandsMen Threads' Salesforce interface. The order number is O-0001, and the owner is Jnanesh Sunkara. The form fields include:

- HandsMen OrderNumber:** O-0001
- Customer:** john
- Status:** Confirmed
- Quantity:** 500 (highlighted in red with an error message: "For Status 'Confirmed', Quantity must be more than 500.")
- Total Amount:** (empty field)
- *Customer email:** jnanesh.sunkara@gmail.com
- Product:** Search HandsMen Products...
- Created By:** Jnanesh Sunkara, 9/12/2026, 4:42 AM
- Owner:** Jnanesh Sunkara

A red error message box is displayed, stating: "We hit a snag. Review the following fields: Quantity". The message also includes the text: "For Status 'Confirmed', Quantity must be more than 500." At the bottom of the form, there are 'Cancel' and 'Save' buttons, with a red prohibition icon next to the 'Save' button.

Figure 23 Order Validation Error

10.1 Automation Test Results

The image displays two screenshots of the HandsMen Threads Salesforce interface. The top screenshot shows the 'HandsMen Product' details for a 'T-shirt'. The bottom screenshot shows the 'HandsMen Order' details for order 'O-0007'. Both screenshots feature a navigation bar with tabs for 'HandsMen Threads', 'HandsMen Customer', 'HandsMen Orders', 'HandsMen Products', 'Inventorys', 'Marketing Campaigns', and 'Rep'. A search bar is located in the top right of each interface.

HandsMen Product Details:

Related	Details
HandsMen Product Name	T-shirt
Order	O-0001
SKU	
Price	\$3
Stock Quantity	4
Created By	Jnanesh Sunkara, 9/12/2026, 4:48 AM
Owner	Jnanesh Sunkara
Last Modified By	Jnanesh Sunkara, 9/12/2026, 6:07 AM

HandsMen Order Details:

Related	Details
HandsMen OrderNumber	O-0007
Customer	john
Status	Confirmed
Quantity	1,000
Total Amount	3,000
Customer email	jnanesh.sunkara@gmail.com
Product	T-shirt
Created By	Jnanesh Sunkara, 9/19/2026, 1:38 AM
Owner	Jnanesh Sunkara
Last Modified By	Jnanesh Sunkara, 9/19/2026, 1:38 AM

Figure 24 Order Total Calculation



Q Search...

HandsMen ThreadsHandsMen Customer ▾HandsMen Orders ▾HandsMen Products ▾Inventories ▾Marketing Campaigns ▾Reps

Inventory

I -0006

Related

Details

Inventory Number

I -0006

Product

[Jeans](#)

Stock Quantity

5,000

Stock Status

Available

Warehouse

Created By

 [Jnanesh Sunkara](#), 9/19/2026, 2:50 AM

Last Modified By

 [Jnanesh Sunkara](#), 9/19/2026, 2:52 AM



Q Search...

HandsMen ThreadsHandsMen Customer ▾HandsMen Orders ▾

✓ HandsMen Order "O-0011" was created. ×

Re

HandsMen Order

O-0011

Related

Details

HandsMen OrderNumber

O-0011

Customer

[john](#)

Status

Confirmed

Quantity

1,000

Total Amount

Customer email

[jnanesh.sunkara@gmail.com](#)

Product

[Jeans](#)

Created By

 [Jnanesh Sunkara](#), 9/19/2026, 2:50 AM

Owner

 [Jnanesh Sunkara](#)

Last Modified By

 [Jnanesh Sunkara](#), 9/19/2026, 2:50 AM



HandsMen Threads
HandsMen Customer
HandsMen Orders
HandsMen Products
Inventories
Marketing Campaigns
Report



Inventory
I -0006

Related
Details

Inventory Number

I -0006

Product

[Jeans](#)

Stock Quantity

4,000

Stock Status

Available

Warehouse

Created By

[Jnanesh Sunkara](#), 9/19/2026, 2:50 AM

Last Modified By

[Jnanesh Sunkara](#), 9/19/2026, 2:50 AM

Figure 25 Inventory Before and After Confirmation

Delete forever
Not spam

3 of 12

Your Order has been Confirmed!
Spam x

Jnanesh Sunkara jnaneshsunkara.ai@gmail.com via [towazrz0tklr.g8-h05kteaz.usa1142.bnc.salesforce.com](#) to me

Sat 12 Sept, 17:13 (7 days ago)

Why is this message in spam? This message is similar to messages that were identified as spam in the past.

Report as not spam

Dear ,
Your order #O-0001 has been confirmed!
Thank you for shopping with us.
Best Regards,
Sales Team

Low stock alert email.
Spam x

Jnanesh Sunkara via [2gyca0sqagrr.g8-h05kteaz.usa1142.bnc.salesforce.com](#) to me

Sat, Sep 12, 5:36 PM (7 days ago)

Why is this message in spam? This message is similar to messages that were identified as spam in the past.

Report as not spam

from: **Jnanesh Sunkara** <jnaneshsunkara.ai@gmail.com>
via [2gyca0sqagrr.g8-h05kteaz.usa1142.bnc.salesforce.com](#)
to: "jnaneshsunkara.ai@gmail.com" <jnaneshsunkara.ai@gmail.com>
date: Sep 12, 2026, 5:36 PM
subject: Low stock alert email.
mailed-by: [2gyca0sqagrr.g8-h05kteaz.usa1142.bnc.salesforce.com](#)
security: Standard encryption (TLS) [Learn more](#)

Dear ,
This is a low stock alert for the product you purchased.
Product name:
Current Stock Quantity: 4
Please take the necessary steps to restock this item immediately.
Best Regards,
Inventory Monitoring System

Low stock alert email. Spam x



Jnanesh Sunkara via vfiy0ywsvqsb.g8-h05kteaz.usa1142.bnc.salesforce.com
to me ▾

Why is this message in spam? This message is similar to messages that were identified as spam in the past.

Report not spam

Dear Inventory Manager,
This is to inform you that the stock for the following product is running low:
Product Name:
Current Stock Quantity: 4
Please take the necessary steps to restock this item immediately.
Best Regards,
Inventory Monitoring System

↩ Reply

➡ Forward

😊

Figure 26 Automated Email Evidence

10.2 Scheduled Automation and Batch Results

HandsMen Customer Name	john		Owner	Jnanesh Sunkara
Email	jnanesh.sunkara@gmail.com			
Phone				
Loyalty Status	Gold			
FirstName	john			
LastName	hamilton			
FullName	john hamilton			
Total Purchases	10,000			
Created By	Jnanesh Sunkara, 9/12/2026, 4:42 AM		Last Modified By	Jnanesh Sunkara, 9/19/2026, 2:59 AM

Figure 27 Loyalty Status Update Result



SETUP

Apex Jobs

[Click here to go to the new batch jobs page](#)

Apex Jobs

Help for this Page

Monitor the status of all Apex jobs, and optionally, abort jobs that are in progress.



Percent of Asynchronous Apex Used: 0%

You have currently used 0 asynchronous Apex operations out of an allowed 24-hour organization limit of 250,000. To learn about how this limit is calculated and what contributes to it, see the [Lightning Platform Apex Limits](#) topic.

View: All [Create New View](#)

Action	Submitted Date	Job Type	Status	Status Detail	Total Batches	Batches Processed	Failures	Submitted By	Completion Date	Apex Class	Apex Method	Apex Job ID
	9/19/2026, 3:16 AM	Batch Apex	Completed		1	1	0	Sunkara Jnanesh	9/19/2026, 3:17 AM	InventoryBatchJob		707g800000i8Owu

Figure 28 Batch Job Execution Result

11 Conclusion

The HandsMen Threads Salesforce project provides a practical CRM and operations solution for a fashion business. It organizes customer, product, order, inventory, and campaign data; applies validation and formula logic; controls access with Salesforce security features; automates customer and stock communication; and uses Apex for business rules and bulk processing.

Together, the configuration, automation, development, and testing evidence demonstrate core Salesforce Platform Developer skills. The completed screenshot set will give the reviewer a clear path from the data model and security design to working automation and functional outcomes.